Parks Meteo Optimization

PEINP Meteorological Data Pipeline - Final Report

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github.com/Pegjoey99/parks-meteo-optimization

Project Overview

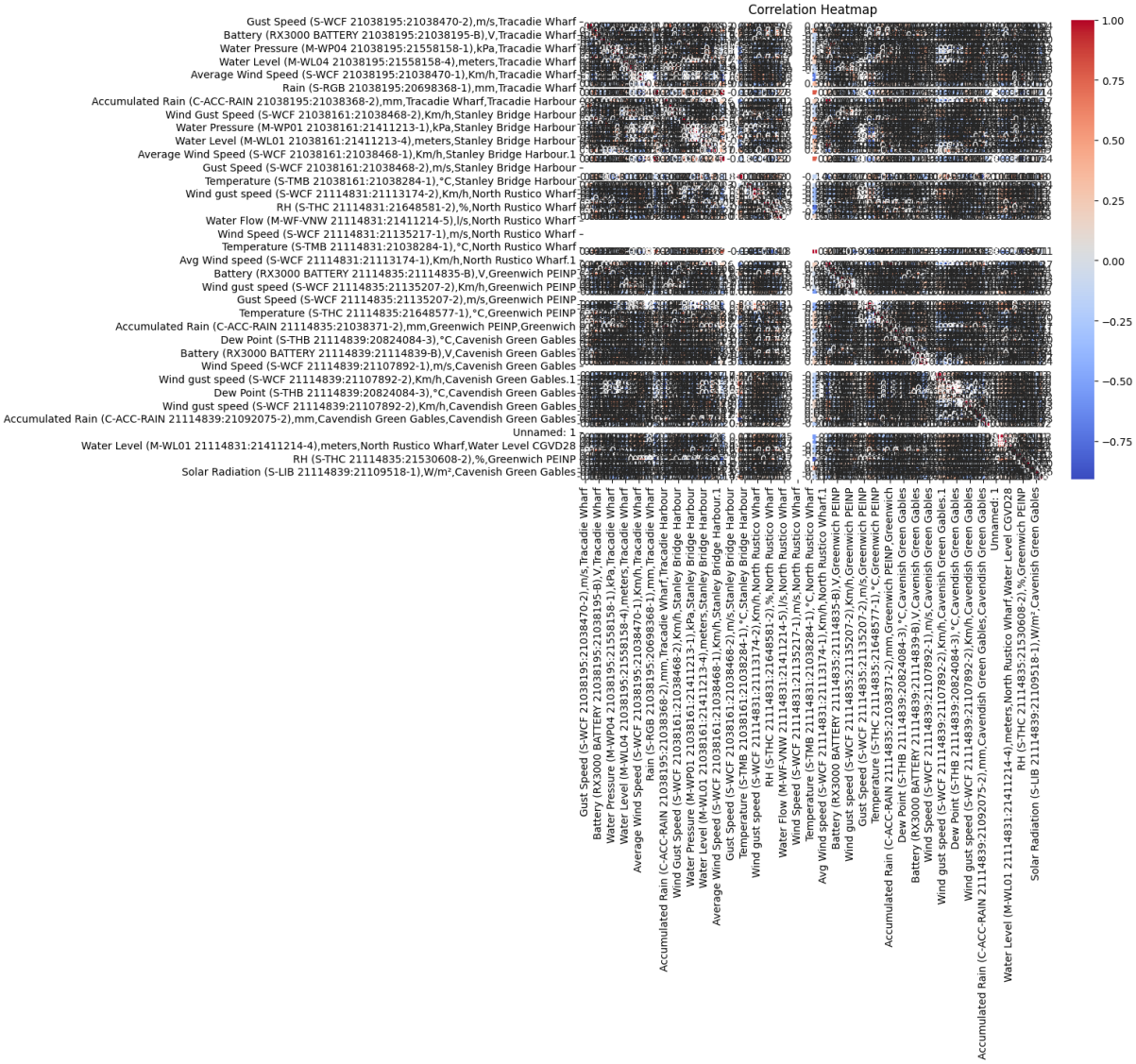
- End-to-end pipeline for meteorological data analysis in Prince Edward Island National Park
- 5 HOBO weather stations: Cavendish, Greenwich, North Rustico Wharf, Stanley Bridge Wharf, Tracadie Wharf
- ECCC Stanhope reference station (Climate ID 8300590) for cross-validation
- Agent-based architecture: Ingest > Clean > Redundancy > FWI > Uncertainty

Data Ingestion & Cleaning

- Ingested 5,086 daily observations across 20 variables
- 100% row retention after cleaning (5,086 > 5,086 rows)
- Column reduction: 20 to 16 columns (dropped redundant/empty fields)
- 2 temperature range issues flagged and handled
- Output: cleaned_data.parquet (380 KB)

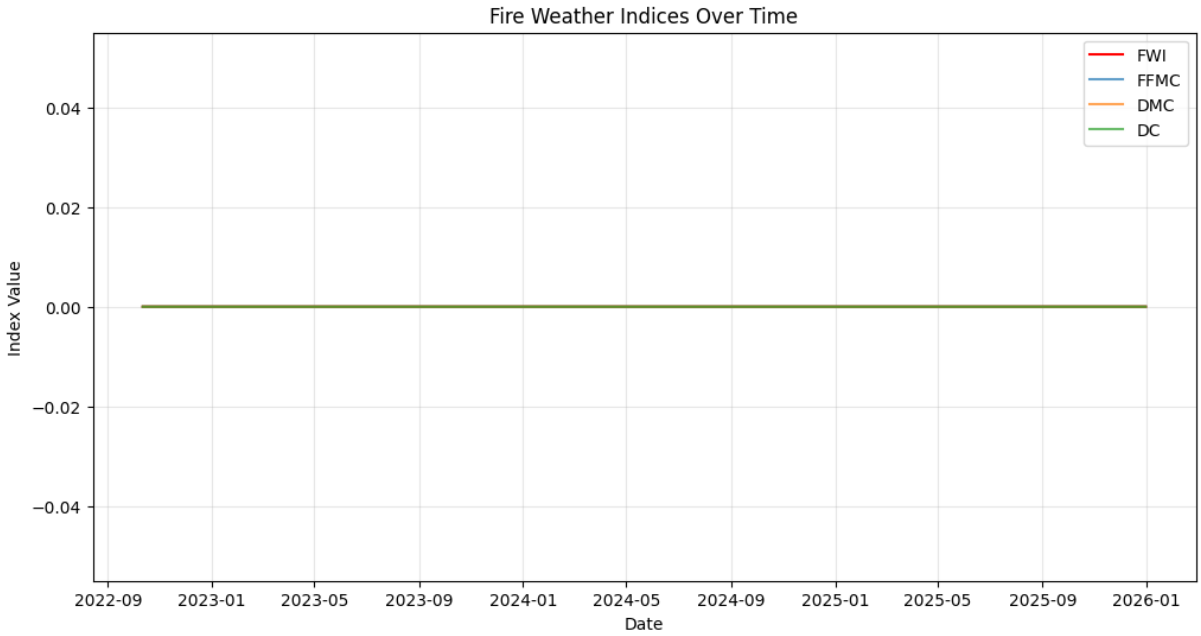
Redundancy Analysis (PCA & Clustering)

- PCA: 4 components capture >90% of variance from 14 sensor channels
- 10 channels carry redundant information - major optimization opportunity
- Wind/gust near-duplicate (r=0.96): decommission gust sensors at peripheral stations
- Pressure variables strongly correlated: shared synoptic weather across park
- 3 K-Means clusters: Cluster 0 (351 extreme days) = all stations active
- Clusters 1-2 (~93% of days): reduced 3-station network sufficient



Fire Weather Index (FWI)

- Van Wagner (1987) Canadian FWI System - 6 indices computed daily
- FWI range: 0.68 - 55.79 (mean 13.06)
- All 6 indices passed physical range validation (PASSED)
- Peak FWI = 55.8 at Stanley Bridge Wharf
- High FWI days (>=10): significant count; Very-High (>=20): flagged for review



FWI Cross-Validation vs Stanhope

- Independent validation against ECCC Stanhope reference (368 fire-season days)
- FFMC MAE = 0.025 - near-identical
- DMC MAE = 0.70, DC MAE = 53.29 (cumulative index, expected divergence)
- ISI MAE = 0.015, BUI MAE = 3.20
- FWI MAE = 0.37 - excellent agreement

Uncertainty & Risk Assessment

- Total Variation (TV) distance: measures cleaning impact on data distributions
- HIGH RISK (>20% shift): water_level (47.9%), water_temperature (42.0%), pressures (36-38%)
- MODERATE RISK (5-20% shift): wind (12.7%), temperature (9.3%), wind_direction (7.4%)
- LOW RISK (<5% shift): rain (1.1%) - critical for FWI Drought Code, highly reliable
- Core FWI inputs (temp, humidity, wind, rain) all <10% shift - FWI calculations trustworthy
- Water sensors need field calibration before 2026 fire season (do not affect FWI directly)

Recommendations for Parks Canada

- 1. DECOMMISSION gust-speed sensors at 2-3 stations (derive from wind, r=0.96, <4% error)
- 2. CONSOLIDATE pressure monitoring to Cavendish (single central sensor replaces 5)
- 3. MAINTAIN all 5 stations for temp, humidity, wind, rain (4 independent FWI inputs)
- 4. SERVICE water_level and water_temperature sensors (42-48% distribution shift = drift)
- 5. CONTINUE full network during fire season (May-Oct); reduced 3-station mode off-season
- Confidence: 5,086 days QC data, PCA, FWI cross-validation (MAE=0.37), KDE uncertainty

Repository & Reproducibility

- GitHub: github.com/Pegjoey99/parks-meteo-optimization
- Jupyter notebook: notebooks/analysis.ipynb - full end-to-end pipeline
- All outputs reproducible via: Run All in VS Code / Jupyter
- Dependencies: pip install -r requirements.txt
- License: MIT